

# Clinical Risk Assessment Report: Patient 11

## 1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 9.7% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.44x the diagnostic threshold

### Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM\_SZE.PET), which reduced the patient's absolute risk by 2.6%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

## 2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

## 3. Protective / Mitigating Factors (Reducing Risk)

**Small Zone Emphasis (PET) (GLZLM\_SZE.PET)** **Value: -0.42 [Avg (29th %ile)] (-2.6%)**

*Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.*

**Patient Age (Age)** **Value: 72.0 [Top 18%] (-1.1%)**

*Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.*

**Long Run High Gray-Level Emphasis (PET) (GLRLM\_LRHGE.PET)** **Value: -0.7 [Bottom 13%] (-1.0%)**

*Points to continuous, elongated bands of high-density or highly active tumor tissue.*

**75th Percentile Tumor Density (CONVENTIONAL\_HUQ3)** **Value: 0.3 [Avg (51th %ile)] (-0.8%)**

*Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.*

**Large Zone Low Gray-Level Emphasis (CT) (GLZLM\_LZLGE.CT)** **Value: -0.41 [Bottom <1%] (-0.7%)**

*Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.*

**Short Run High Gray-Level Emphasis (CT) (GLRLM\_SRHGE.CT)** **Value: 0.17 [Avg (47th %ile)] (-0.7%)**

*Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.*

**Squamous Cell Carcinoma Antigen (SCC\_Ag)** **Value: 2.2 [Top <1%] (-0.6%)**

*Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.*

**PET Radiomic Metabolic Feature (GLZLM\_ZLNU.PET)** **Value: -0.49 [Bottom <1%] (-0.4%)**

*Metabolic PET feature reflecting glucose avidity and cellular metabolic rate.*

Machine Learning Feature Attribution (SHAP Decision Path)

